

Establishing the identity of Anna Anderson Manahan

Sir — The 70-year controversy surrounding the identity of Anna Anderson Manahan has been remarkable in capturing the imagination of so many people. Supporters of her claim said that she was the Duchess Anastasia, daughter of Tsar Nicholas II and had dramatically escaped execution by a Bolshevik firing squad on July 16th, 1918. Disbelievers denied the claim, but could not establish her true identity.

The recent excavation of a mass grave at Ekaterinburg in the Urals of Central Russia discovered the remains of nine bodies. DNA profiles¹ from bones were compared to blood samples from living relatives, including HRH the Duke of Edinburgh establishing that the remains were those of the Russian Royal family. Intriguingly, two bodies were missing — those of Prince Alexei and a princess. DNA profiling could not establish the identity of the missing princess, perpetuating the mystery for those who believed that Anastasia had survived.

One theory put forward by disbelievers was that Anna Anderson was in fact Franziska Schanzkowska. This was first proposed by a private detective hired by the Grand Duke of Hesse in the 1920s. However, despite intensive investigation, remarkably little is known about the life of Schanzkowska prior to 1920. She was born in approximately 1896 and lived in Pomerania in the north of Germany, adjacent to the Polish border. During the first world war, working in a munitions factory in Berlin she was badly injured in an explosion. Subsequently, she was admitted to two different mental hospitals, but disappeared in 1920.

About the same time, Anna Anderson appeared and claimed to be the Royal Duchess Anastasia.

Following the Forensic Science Service (FSS) work last year¹ to identify the remains of the Tsar and his family, the FSS and the Armed Forces Institute of Pathology (AFIP) were asked to carry out independent analyses on small bowel samples removed from Anna Anderson Manahan during the course of surgery at the Martha Jefferson Hospital in Charlottesville, Virginia in 1979. The tissue samples had been fixed in formaldehyde and preserved in paraffin wax. In all there were five different blocks, two of which were analysed. The blocks were identified by serial numbers which corresponded to notes made in the hospital records. Both testing laboratories were allowed to take several 6 µm microtome slices from each block. In addition, six hairs said to have come from Anna Anderson were analysed by the Department of Anthropology, Pennsylvania State University.

DNA was extracted from paraffin wax blocks using the same protocol².

The extracted DNA was found to be highly degraded (less than 250 bp in size). The FSS carried out short tandem repeat (STR) analysis using a quadruplex³ and mtDNA analysis using a modification of a solid phase sequencing protocol⁴. AFIP performed mitochondrial (mt) DNA analysis using dye-terminator cycle sequencing⁵. Hair extraction was carried out by Pennsylvania State University following the method of Stoneking and Melton⁶.

We compared five STRs from two different paraffin wax blocks to the

DNA profiles analysed from bones of the Tsar and Tsarina (Table 1); four out of five different STRs were inconsistent with the hypothesis of parentage. In addition, we carried out sex typing of the tissue using the amelogenin test⁷.

The non-coding region of mtDNA was sequenced and compared to a blood sample donated by HRH the Duke of Edinburgh, a direct maternal descendant of the Tsarina. We found six differences in hypervariable region 1 (Table 2), which again disproved the maternal relationship theory.

In addition to comparing DNA profiles from the tissue samples with living maternal relatives of the Russian Royal family, we also analysed blood donated by Carl Maucher, the great nephew of Schanzkowska and directly related through the maternal line, to test the possibility of relationship. A match was observed between the DNA samples at the positions indicated in Table 2. This suggested that Carl Maucher may be a maternal relative of Anna Anderson.

In conclusion, two different samples (hair and intestine) said to have come from Anna Anderson have been independently collected and analysed by three different laboratories. Because our results are in agreement, this strongly suggests that the samples came from the same individual (Anna Anderson herself); the contention that the samples came from another (unknown) individual seems highly unlikely. The samples could not be associated with a maternal relative of the Tsarina or of HRH Prince Philip.

We have considered two possibilities: (i) that Anna Anderson and Carl Maucher are maternally related;

	VWA	TH01	F13A1	FES/FPS	ACTBP2	AMELOGENIN
Tsar (Skeleton 4) ^a	15,16	7,9,3 ^b	7,7	12,12	11,32	X,Y
Tsarina (Skeleton 7) ^a	15,16	8,8	3,2 ^c ,5	12,13	32,36	X,X
Anna Anderson (intestine sample)	14,16	7,9,3	3,2,7	11,12	15,18	X,X

Table 2 Mitochondrial DNA sequences

Origin of sample	DNA source	Length sequenced (bp)	Position within the non-coding region (ref. 11)					
			16111	16126	16266	16294	16304	16357
Reference sequence	—	—	C	T	C	C	T	T
Duke of Edinburgh (Great nephew of Tsarina)	Blood sample	403	T	.	.	.	.	C
Anna Anderson	Intestine sample	403	.	C	T	T	C	.
Anna Anderson	Hair sample	344–362 (3 hairs)	.	C	T	T	C	.
C. Maucher (Great nephew of Schanzkowska)	Blood sample	380	.	C	T	T	C	.

., Sequence unchanged from reference sequence.

(ii) that they are unrelated. To assess the strength of the evidence we compared the DNA profile from the Maucher and putative Anna Anderson samples with over 300 Caucasian sequences in published^{8–10} and unpublished databases. We did not find a match, which suggests that the DNA sequence is rare. Assuming that the databases we have used are

representative of European Caucasians, the chance of finding matching DNA profiles if Carl Maucher and Anna Anderson are unrelated through the maternal line is less than one in 300. This finding supports the hypothesis that Anna Anderson and Franzisca Schanzkowska were the same person.

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Polymerase δ variants in RER colorectal tumours

Sir — The replication error (RER) phenotype is characterized by genetic instability, particularly in microsatellite sequences, and occurs in both sporadic and hereditary forms of colorectal cancer^{1–3}. In the hereditary form, mutations of mismatch repair (MMR) genes are usually responsible for the instability. In many sporadic tumors, however, mutations of the four known MMR genes are apparently absent (see pp. XX–YY, this issue)⁴. We have therefore begun to investigate whether

other genes involved in replication or repair might contribute to the RER phenotype.

Mutator phenotypes have been extensively studied in unicellular organisms such as *E. coli* and *S. cerevisiae*, in which mutations in the 3'–5' exonuclease 'proofreading' domain of DNA polymerases can cause a mutator phenotype similar to that observed in RER cancers. Interestingly, mutations affecting the exonuclease activity associated with

E. coli pol III or *S. cerevisiae* pol δ can cause a mismatch repair deficiency or microsatellite instability^{5,6}, prompting us to study the exonuclease domain of human pol δ .

We have studied eight colorectal cancer cell lines which displayed microsatellite instability^{4,7} and were negative for mutations of the four known human MMR genes. All cell lines (HCT15, DLD-1, LS180, RKO, 587X, VACO 444, VACO 457 and VACO 481) were examined for MMR gene mutations by reverse transcriptase — polymerase chain reaction (RT-PCR) analysis to detect abnormal transcripts and by the *in vitro* synthesized protein assay to detect abnormal polypeptides. These